Taylor Series Worksheet — Worked Answer Key

Twenty problems on known series, transformations, coefficients, intervals, approximations, and error.

Revision: 2026-07-23 **Canonical page:** https://bettergrades.net/subjects/math/calculus/worksheets/taylor-series/

1. Find the degree-3 Maclaurin polynomial for e to the power x.

   Answer: $1 + x + x^2/2 + x^3/6$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $1 + x + x^2/2 + x^3/6$.

2. Find the degree-5 Maclaurin polynomial for sin x.

   Answer: $x - x^3/6 + x^5/120$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $x - x^3/6 + x^5/120$.

3. Find the degree-4 Maclaurin polynomial for cos x.

   Answer: $1 - x^2/2 + x^4/24$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $1 - x^2/2 + x^4/24$.

4. Write the first four nonzero terms of ln(1+x).

   Answer: $x - x^2/2 + x^3/3 - x^4/4$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $x - x^2/2 + x^3/3 - x^4/4$.

5. Expand 1/(1-x) as a power series.

   Answer: $\sum_{n=0}^{\infty} x^n,\ |x| < 1$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $\sum_{n=0}^{\infty} x^n,\ |x| < 1$.

6. Expand 1/(1+x to the power 2) as a power series.

   Answer: $\sum_{n=0}^{\infty} (-1)^n x^{2n},\ |x| < 1$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $\sum_{n=0}^{\infty}(-1)^n x^{2n},\ |x| < 1$.

7. Expand 1/(1-3x) as a power series.

   Answer: $\sum_{n=0}^{\infty} 3^n x^n,\ |x| < 1/3$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $\sum_{n=0}^{\infty} 3^n x^n,\ |x| < 1/3$.

8. Find a power series for 1/(1-x) to the power 2.

   Answer: $\sum_{n=1}^{\infty} nx^{n-1},\ |x| < 1$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $\sum_{n=1}^{\infty} nx^{n-1},\ |x| < 1$.

9. Find a power series for arctan x by integrating 1/(1+x squared).

   Answer: $\sum_{n=0}^{\infty}(-1)^n x^{2n+1}/(2n+1)$

   a. Start from derivatives at the center or a known convergent power series.
   b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
   c. The resulting expression is $\sum_{n=0}^{\infty}(-1)^n x^{2n+1}/(2n+1)$.

10. Find the degree-2 Taylor polynomial for ln x centered at 1.

    Answer: $(x-1) - (x-1)^2/2$

    a. Start from derivatives at the center or a known convergent power series.
    b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
    c. The resulting expression is $(x-1) - (x-1)^2/2$.

11. Write the Taylor series for e to the power x centered at a.

    Answer: $e^a \sum_{n=0}^{\infty}(x-a)^n/n!$

    a. Start from derivatives at the center or a known convergent power series.
    b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
    c. The resulting expression is $e^a \sum_{n=0}^{\infty}(x-a)^n/n!$.

12. If f to the power (n)(0)=2 to the power n, find the Maclaurin series.

    Answer: $\sum_{n=0}^{\infty} 2^n x^n/n! = e^{2x}$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $\sum_{n=0}^{\infty} 2^n x^n/n! = e^{2x}$.

13. Use T_3 for e to the power x to approximate e to the power (0.1).

Answer: $1.105166\overline{6}$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $1.105166\overline{6}$.

14. Use x-x cubed/6 to approximate sin(0.2).

Answer: $0.198666\overline{6}$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $0.198666\overline{6}$.

15. Find the radius of $\sum nx^n/4^n$.

Answer: $R = 4$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $R = 4$.

16. Find the interval of convergence of $\sum x^n/n$.

Answer: $[-1, 1)$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $[-1, 1)$.

17. Find the interval of convergence of $\sum(x-2)^n/(n3^n)$.

Answer: $[-1, 5)$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $[-1, 5)$.

18. Find R for $\sum n!(x+1)^n/(2n)!$.

Answer: $R = \infty$

a. Start from derivatives at the center or a known convergent power series.
b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
c. The resulting expression is $R = \infty$.

19. Bound the error of the degree-3 Maclaurin approximation to e to the power (0.2).

    Answer: $|R_3| \leq e^{0.2}(0.2)^4/4! < 0.000082$

    a. Start from derivatives at the center or a known convergent power series.
    b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
    c. The resulting expression is $|R_3| \leq e^{0.2}(0.2)^4/4! < 0.000082$.

20. How many terms of the alternating arctan series ensure error below 0.001 at x=1/2?

    Answer: 4 terms

    a. Start from derivatives at the center or a known convergent power series.
    b. Apply substitution, differentiation, integration, or the ratio test while preserving the convergence condition.
    c. The resulting expression is 4 terms.